

Optimal Augmentation of Distribution Networks for Improved Reliability

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Abstract—Reliability is a crucial factor influencing electricity network planning and has gained considerable attention in the past decade due to the increasing number of weather-related outage events. This article explores connections between network topology and reliability, and proposes a graph-based approach to enhance the reliability of distribution systems by installing new tie-lines. The problem is cast here as an edge-connectivity augmentation problem, with 2-edge connectivity being desired to ensure service restoration under the outage of any single line in the network. However, in practice, such a target may not be economical or feasible. Therefore, a systematic optimization framework is presented in this article to model partial augmentations to the system for improving the reliability at a select subset of nodes. This approach allows for selectively strengthening parts of the network based on connectivity requirements. The article also proposes the use of graph-based metrics to characterize the reliability of distribution networks. Applications of the approach to realistic distribution test systems demonstrate its effectiveness in identifying line additions that can favorably impact system reliability. Numerical results also quantify the reliability improvement achievable through partial augmentation.

Index Terms—Distribution system planning, graph algorithms, optimization, reliability.

NOMENCLATURE

G	The network graph.
V	Set of vertices in G .
E	Set of edges in G .
E'	Set with edges of E in reversed orientation.
N	Number of nodes in G .
$\bar{G}$	Complement graph of G .
$\kappa_e(G)$	Global edge-connectivity of G .
$\kappa_e(i, j)$	Local edge-connectivity between nodes i and j .
$\bar{\kappa}_e(G)$	Average edge-connectivity of G .
$t(G)$	Number of spanning trees of G .
V^c	Set of critical nodes.
$k(t)$	Edge-connectivity requirement at node t .
G_0	Initial graph with vertex-set V and edge-set E_0 .
E_0	Set of edges in the initial graph G_0 .
E_x	Candidate edges for augmentation.
x_{ij}	Decision variable representing the status of edge (i, j) .

w_{ij}	Weight associated with the edge (i, j) .
E_*	Set of edges in the optimal augmentation.
$y_{ij}^{(st)p}$	Variable capturing the membership of line (i, j) in the p th path between nodes s and t .
$c_{eB}(e)$	Edge-betweenness centrality of the edge e .
$\Delta t(G_e)$	Increase in the number of spanning trees due to the addition of edge e to G .

I. INTRODUCTION

RELIABILITY has traditionally played a pivotal role in power system planning and operations. Recent articles indicate declining trends in distribution reliability due to severe weather-related interruptions [1], warranting the need to explicitly include reliability considerations into distribution network planning [2]. The reliability of existing distribution networks can be improved by adding protection devices like fuses and reclosers, installing new switching devices, automating operations, improving crew response times, and augmenting the network by installing new tie-lines and feeders [3]. This article presents a formulation for distribution reliability enhancement via the installation of new tie-lines, using graph algorithms and optimization.

Distribution systems are often operated radially. In the absence of normally open tie-switches, such a network cannot survive the failure of even a single component. Thus, distribution networks typically have some tie-switches which are kept open under normal operations but can be closed under line or transformer failures to connect load points to the substation through alternative paths. This improves system reliability by minimizing the duration of customer interruptions.

The installation and optimal placement of new switching devices as a means to improve reliability have received considerable attention in literature; see the survey [4] and references therein. For instance, optimizing the number and locations of switches [5] and replacing manual switches with remote-controlled switches [6] are commonly reported approaches for improving reliability. Junlakarn and Ilić [7] propose the installation of switches on existing lines along with the addition of new tie-lines to facilitate service restoration. The approaches in [5] and [6] do not alter the network graph, while the line additions considered in [7] manifest as topology changes. While it is understood that network topology has a significant impact on reliability, a systematic framework to study this does not exist. Hence, the proposed planning approach in this article only looks at the addition of new lines to impact reliability.

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Literature focusing on only new line additions for reliability enhancement is limited. The expansion planning problem presented in [8] addresses the installation of new sectionalizers and tie-lines. Tie-lines are identified using a multiobjective particle swarm optimization algorithm. Reliability is factored in as the average load not served by considering the failure of one line at a time. The solution scheme proposed in [8] searches over all possible tie-lines. For large networks, the search-space can be quite large, and the possibility of the heuristic algorithm getting stuck in suboptimal solutions is high. The work in [9] incorporates reliability by setting limits on reliability indices. However, their solution scheme employing heuristic search and simulated annealing is designed for the construction of new distribution networks instead of augmentation.

The approach in [10] considers the installation of new lines with reliability improvement as one of the many objectives. However, the results presented utilize new lines only for serving new load-points. The work in [11] also considers the installation of tie-lines along with substations and distributed generation. Reliability is modeled as the cost of energy not supplied and is included in the objective function of the optimization problem. The optimization formulation is then solved using a genetic algorithm. Since the planning problem integrates an array of options for network expansion, it is difficult to analyze the sensitivity of any one factor on the reliability of the network.

Muñoz-Delgado *et al.* [12] present an optimization framework for network expansion that explicitly includes expected energy not supplied as a reliability metric. The solution of the multistage optimization provides the locations of new substations and lines that minimize the investment and operation costs. The optimal solution on a 54-node test system offers lower overall costs when compared to alternatives obtained using heuristic methods. The reliability quantification, however, ignores the presence of tie-lines and post-fault reconfiguration. Another optimization framework that considers reliability enhancement through the addition of tie-lines can be found in [13]. Tie-line addition is considered along with the installation of new switches on existing lines. Reliability is part of the objective function and is modeled as a function of standard reliability metrics. In the model presented, the locations of tie-lines are limited to the ends of radial feeders which may lead to suboptimal solutions in large networks.

The planning methodology in [14] uses graph theory notions of spanning trees and cycles to find optimal topologies that minimize investment costs, reduce power losses, and improve reliability by reducing interruption costs. An evolutionary optimization searches over all spanning trees of the network graph to find minimum-cost topologies. The algorithm also identifies circuit backups (tie-lines) that improve reliability but are economical to invest in.

Graph-theoretic approaches to improving system reliability are popular in communication networks under the class of problems known as *design of survivable networks* [15]. Xu *et al.* [16] propose a method to design reliable communication networks by adding normally inactive replacement links while the primary network is operated radially. The network design problem is cast as an *edge-connectivity augmentation*

(ECA) problem in [16]. Since the replacement links are analogous to the normally open tie-lines in power distribution networks, this article draws inspiration from [16] to use the ECA framework to address the tie-line addition problem of interest.

A 2-edge-connected distribution network will have service restoration options available under all single line outages. Therefore, 2-edge-connecting a network is equivalent to making it $N - 1$ secure against line outages. The mixed-integer linear programming (MILP) formulation for distribution network expansion proposed in [17] attempts to explicitly model the $N - 1$ security criterion while optimizing the addition of new feeders and substations. Singh *et al.* [18] propose a model where the notion of $N - 1$ security is modeled using a two-stage stochastic mixed-integer programming formulation. The addition of new feeders, transformers, and substations is considered. However, the complexity of the solution approach can grow quickly as more failure scenarios are considered.

A previous work by Jose and Kowli [19] uses ECA algorithms from graph theory to bring in the notion of $N - 1$ security in distribution system planning. This article extends the preliminary approach proposed in [19] by systematically bringing practical considerations while also presenting an optimization framework for ECA. Indeed, the integer-linear programming (ILP) formulation proposed here is a key contribution of this article. Its superiority in addressing the distribution system planning problem for reliability enhancement over the graph algorithms used in [19] is demonstrated via several case studies.

Completely 2-edge-connecting a distribution system is often impractical, unnecessary, and uneconomical. An approach that selectively improves the connectivity of certain critical nodes in the network is more relevant from the utility planner's point of view. This problem is a *local ECA* problem, for which existing graph algorithms typically give approximate solutions. The proposed ILP scheme leverages the characteristics of distribution networks to arrive at globally optimal solutions for the local ECA problem. To the best of the authors' knowledge, this is a novel contribution for improving the local connectivity of networks in a tractable manner.

A related approach for reliability enhancement in distribution networks is the distribution feeder reconfiguration problem. However, it is to be noted that reconfiguration is only possible in the presence of tie-lines. The tie-line addition problem tackled in this article can be used to provide flexibility for optimal reconfiguration and service restoration. This is certainly relevant for networks with zero or a limited number of tie-lines, as may be the case with the test systems used for the case studies in this article.

The impact of network topology on distribution reliability indices like SAIFI, SAIDI, and EENS is established in existing literature [20]. In addition to the topology, the specific values of these indices are functions of component failure rates, number of customers, load patterns, and service restoration strategy of the utility. Hence, it is difficult to establish a precise relationship between network topology, its augmentation, and the improvement in standard reliability indices. Approaches for incorporating standard reliability indices in network optimization—see [21] for an example—rely on making several simplifying

assumptions about the failure probabilities and service restoration procedure. To circumvent this issue, an attempt has been made to quantify distribution reliability as a function of topology alone using graph-based metrics. However, empirical results obtained in [22] indicate that ECA favorably impacts standard reliability metrics.

The scope of the article is limited to identifying the number and locations of new tie-lines to be installed. The decisions on the capacity of the new lines and operational considerations like voltage drops are left to a second-stage planning study. In this context, the proposed approach does not offer a complete expansion planning solution. Instead, the optimal solutions can serve as candidate lines in the traditional planning schemes reviewed above and in operational planning such as [23].

In summary, the major contributions of this article are as follows.

- A modeling framework that casts distribution network expansion for reliability improvement as a graph augmentation problem.
- An ILP optimization formulation that solves the global and local variants of the 2-edge-connectivity augmentation problem. The local variant allows for selectively improving the reliability of certain parts of the network.
- Approaches to incorporate practical considerations arising from engineering, economic, and policy factors in the optimization formulations.
- A set of graph-based metrics that relate network reliability and topology.

The rest of the article is organized as follows. Section II introduces the ECA problem and the reliability metrics. The proposed ILP formulation is discussed in Section III. Section IV addresses certain practical considerations in using ECA for distribution networks. Numerical results are analyzed in Section V, and concluding remarks are included at the end.

II. EDGE-CONNECTIVITY AUGMENTATION

This section introduces the ECA problem and its variants that may be used for improving the reliability of distribution networks. To simplify reliability characterization, standard assumptions used in feeder reconfiguration and distribution reliability studies are invoked: the network is balanced, component failures are independent, and all lines are equipped with switching devices.

A. Network Graphs and Edge-Connectivity

The topology of a power distribution network can be represented using an undirected graph $G = (V, E)$, where the vertex-set V corresponds to the nodes, and the edge-set E corresponds to the lines (including normally open tie-lines). An edge between nodes i and j is denoted as (i, j) . Parallel lines between nodes i and j in the network, if any, will be represented in G using an equivalent single edge. Despite G being undirected, assuming that the edge (i, j) is directed from i to j will simplify the optimization formulation in Section III. With this assumption, let $E' = \{(j, i) \mid (i, j) \in E\}$ denote the edge-set with reversed edges. For the problems considered in this article, the edges are

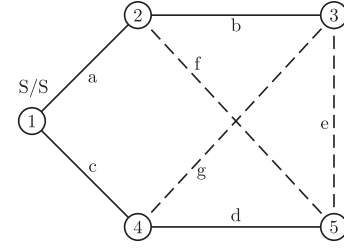


Fig. 1. Graph of the example system. Edges labeled e , f , and g are in $\bar{G}$.

allowed to have non-negative weights w_{ij} corresponding to each edge $(i, j) \in E$. The complement of the graph G , denoted as $\bar{G}$, is the graph having the same vertex set V as G and with all the edges in $V \times V$ which are not present in E .

A *path* between two nodes v_1 and v_k is a sequence of nodes $\{v_1, \dots, v_k\}$ such that the edges $(v_i, v_{i+1}) \in E, \forall i \in \{1, \dots, k-1\}$. The path can also be described using the sequence of edges connecting v_1 and v_k . A graph is said to be connected if there is a path between every pair of its nodes.

The level of connectivity of a graph can be quantified using the notion of *edge-connectivity*. The edge-connectivity $\kappa_e(G)$ of the connected graph G is the minimum number of edges whose removal either disconnects G or reduces it to a single-node graph [24]. A graph is called k -edge-connected if $\kappa_e(G) \geq k$. Edge-connectivity can also be defined between any pair of nodes in G . Termed as *local edge-connectivity* and denoted as $\kappa_e(i, j)$, this metric denotes the minimum number of edges whose removal disconnects i and j . Using this definition, a k -edge-connected graph is a special case where $\kappa_e(i, j) \geq k, \forall (i, j) \in V \times V, i \neq j$.

The concept of edge-connectivity can be illustrated using the example system shown in Fig. 1. For convenience, the edges are labeled using lower case alphabets. Consider the graphs $G = (V, E)$ with $V = \{1, 2, 3, 4, 5\}$ and $E = \{a, b, c, d\}$, $G_e = (V, E \cup \{e\})$, and $G_f = (V, E \cup \{f\})$. Clearly, $\kappa_e(G) = 1$ since the removal of any one edge disconnects G . Likewise, $\kappa_e(G_f) = 1$ since removal of b disconnects the graph. On the other hand, $\kappa_e(G_e) = 2$ since at least two edges have to be removed from G_e to disconnect it.

A *bridge* is an edge whose removal disconnects the graph. In the previous example, b is a bridge in the graph G_f . The graph G_e is bridgeless. It is easy to verify that a connected, bridgeless graph will be 2-edge-connected.

B. Quantifying Reliability as a Function of Topology

Since the focus of this article is on the effect of topology on reliability, the following metrics are introduced as indicators of the reliability of a given network topology.

1) *Average Edge-Connectivity*, $\bar{\kappa}_e(G)$: Global edge-connectivity is a worst-case metric. For example, in Fig. 1, adding either the edge f or g improves the connectivity of four out of the five nodes. But the edge-connectivity of the graph remains as *one*, owing to the one node having only a single path to the substation node. Average edge-connectivity is defined to

capture the overall reliability as

$$\bar{\kappa}_e(G) = \frac{\sum_{t \in V(G), t \neq s} \kappa_e(s, t)}{N - 1} \quad (1)$$

where s is the substation (source) node and N is the number of nodes in G . Using (1), the $\bar{\kappa}_e$ of the example in Fig. 1 with the addition of edge f alone is calculated as $\bar{\kappa}_e(G_f) = 1.75$. The $\bar{\kappa}_e$ of a globally 2-edge-connected graph is 2.

2) *Number of Spanning Trees, $t(G)$* : The motivation for using $t(G)$ as a reliability indicator comes from the fact that regardless of the objective function (of a feeder reconfiguration or service restoration strategy), any optimal configuration of a radially operated distribution network will be a spanning tree of its graph G . In the context of distribution networks, the higher number of spanning trees implies more options for service restoration and increased overall reliability. The value of $t(G)$ can be calculated using *Kirchoff's matrix tree theorem* [25].

C. Edge-Connectivity Augmentation

The 2-edge-connectivity augmentation problem attempts to add a minimum weight set of edges to G to make it 2-edge-connected. For instance, the graph G in Fig. 1 can be 2-edge-connected by adding edge e , while addition of edge f alone does not improve its overall edge-connectivity. An important consequence of k -edge connecting a graph is that it introduces k edge-disjoint paths between any pair of vertices in the graph. This follows from Menger's theorem [26] for undirected graphs. Edge-disjoint paths do not share an edge.

In distribution systems, we are interested in the local edge-connectivity of nodes with a substation, which will be designated as the source node. 2-edge-connectivity for a network would imply the existence of two edge-disjoint paths to the source node. This, in turn, would allow service restoration via network reconfiguration under all single component outages. In the remainder of the article, when the local edge-connectivity of a node is mentioned, the implication is that the metric is evaluated between that node and the source node. When a graph is globally 2-edge-connected, every pair of nodes will have at least two edge-disjoint paths between them. In such cases, the notion of a source node is redundant. However, this article will continue to refer to both global and local connectivity with respect to the source node without any loss of generality.

Local edge-connectivity augmentation can be used to selectively improve the reliability of parts of the network. For the network in Fig. 1, the graph G_f gives a local edge-connectivity of *two* for all nodes except *node 3*. Similarly, choosing the edge g will 2-edge-connect all nodes except *node 5*. The metrics introduced in Section II-B are suitable for quantifying the reliability enhancement even with such partial augmentation. The following example illustrates this.

Example: Consider the example system of Fig. 1, where $t(G) = 1$ and $\bar{\kappa}_e(G) = 1$. Despite being 1-edge-connected, G_f with $t(G_f) = 4$ and $\bar{\kappa}_e(G_f) = 1.75$ offers higher reliability to the loads at nodes 2, 4, and 5 as compared to G . On the other hand, the graph G_e is 2-edge-connected (with $t(G_e) = 5$ and $\bar{\kappa}_e(G_e) = 2$) and offers improved reliability to all nodes.

D. Graph Algorithms for ECA

There are efficient graph algorithms available for solving the global 2-edge-connectivity augmentation problem. Two specific variants are considered in this article.

- 1) *Global Unweighted Edge-Connectivity Augmentation (GU-Graph)*: The edges are unweighted (or have equal weights) and all edges in $\bar{G}$ are available for augmentation. Under these conditions, the objective of the augmentation problem reduces to finding a minimum number of edges to be added to 2-edge connect G .
- 2) *Global Weighted Edge-Connectivity Augmentation (GW-Graph)*: In this problem, edges with finite, positive weights are to be added from a given subset of edges in $\bar{G}$.

For the unweighted variant, an optimal solution can be obtained using the algorithm outlined in [27]. The algorithm presented in [28] for the weighted problem achieves a 2-approximation of the solution.

Despite being fast and scalable, these graph algorithms have certain limitations in their application to distribution network augmentation. Since the unweighted algorithm selects edges from the complete graph, some of the lines in the optimal augmentation might be technically or economically not viable. One way to incorporate such considerations in the augmentation problem is to associate weights to the candidate edges based on their installation costs. While the weighted version can be applied to this problem, the algorithm can only guarantee an approximate solution, which can be sub-optimal for distribution systems.

If higher reliability is desired only at a subset of *critical nodes*, this problem can be modeled as a local ECA problem. Approaches available in the literature for solving the local weighted ECA problem invoke several assumptions about the underlying network and the edge-weights [29], [30], making them not suitable for distribution network augmentation. To develop a tractable and extensible algorithm, this article formulates and uses the ILP version described in Section III.

III. OPTIMIZATION FORMULATION

This section presents an ILP formulation for the local k -edge-connectivity augmentation, which can also be suitably modified to solve the global edge-connectivity augmentation problem.

A. Optimal Edge-Connectivity Augmentation

Using Menger's theorem, the augmentation problem for k -edge-connecting a pair of nodes s and t can be cast as a minimum-cost edge selection problem, which ensures the existence of at least k edge-disjoint paths between s and t in G . Since the connectivity of nodes with respect to the substation is of interest, the substation is always designated as the source node s . The formulation can be extended to handle the presence of multiple substations by merging all substation nodes into a single source node.

If the set of critical nodes in G is represented as V^c where $V^c \subseteq V$, then the connectivity requirements for the nodes

$t \in V \setminus s$ with respect to s may be assigned using a function $k(t)$ defined as

$$k(t) = \begin{cases} 2, & \text{if } t \in V^c \\ 1, & \text{otherwise.} \end{cases} \quad (2)$$

Let $G_0 = (V, E_0)$ denote the graph corresponding to the initial topology of the distribution network. Let E_x represent the candidate edges in $\bar{G}$ which are available for augmentation. The optimization is performed over the super-graph $G = (V, E)$ where $E = E_0 \cup E_x$. The problem can be formulated as an ILP by using binary variables x_{ij} to represent the status of edges where

$$x_{ij} = \begin{cases} 1, & \text{if } (i, j) \in E \text{ is selected in the solution} \\ 0, & \text{otherwise.} \end{cases} \quad (3)$$

Associated with each edge $(i, j) \in E$ is a weight w_{ij} that models the cost of selecting the edge (i, j) . Let $E_* \subseteq E$ defined as

$$E_* = \{(i, j) \mid x_{ij} = 1, (i, j) \in E\} \quad (4)$$

denote the optimal set of edges obtained as a solution of the optimization problem. Since the problem seeks an optimal augmentation, the formulation must ensure that $E_0 \subseteq E_*$. Equality occurs if the initial topology already satisfies the connectivity requirements. The proposed ILP formulation is as follows:

$$\min \sum_{(i,j) \in E} w_{ij} x_{ij} \quad (5)$$

subject to

$$x_{ij} = 1, \quad \forall (i, j) \in E_0 \quad (6)$$

$$\sum_{j:(i,j) \in E \cup E'} y_{ij}^{(st)p} - \sum_{j:(j,i) \in E \cup E'} y_{ji}^{(st)p} = f(i) \quad \forall t \in V \setminus s, p = 1, \dots, k(t), i \in V \quad (7)$$

where

$$f(i) = \begin{cases} 1, & \text{if } i = s, \\ -1, & \text{if } i = t, \\ 0, & \text{if } i \in V \setminus \{s, t\} \end{cases} \quad \forall i \in V$$

$$\sum_{p=1}^{k(t)} \left(y_{ij}^{(st)p} + y_{ji}^{(st)p} \right) \leq x_{ij} \quad \forall t \in V \setminus s, (i, j) \in E \quad (8)$$

$$\sum_{j:(i,j) \in E} x_{ij} + \sum_{j:(j,i) \in E} x_{ij} \geq k(i) \quad \forall i \in V \quad (9)$$

$$x_{ij} \in \{0, 1\} \quad \forall (i, j) \in E$$

$$y_{ij}^{(st)p} \in \{0, 1\} \quad \forall (i, j) \in E \cup E'. \quad (10)$$

The objective function (5) represents the total cost of adding the new tie-lines. Constraint (6) ensures that the condition $E_0 \subseteq E_*$ is satisfied by the solution. In the absence of (6), the formulation solves a graph synthesis problem since it makes no distinction

between the edges already present in the graph and the candidate edges.

Constraint (7) indicates the existence of $k(t)$ paths between s and every $t \in V \setminus s$. The binary variable $y_{ij}^{(st)p}$ denotes whether the edge $(i, j) \in E \cup E'$ is selected in the p th path from s to t or not. Constraint (7) ensures that there is one edge directed away from s , one edge terminating at t , exactly two edges incident on any node in the path other than s and t , and no edge incident on nodes not visited by the path.

Constraint (8) ensures that the $k(t)$ paths between a given (s, t) pair are edge-disjoint. The minimum degree of a node t is at least $k(t)$, which is captured in constraint (9). While this constraint is not necessary for solving the problem, it helps in reducing the search-space. The formulation can be used to solve the global augmentation problem by defining $k(t) = k, \forall t \in V \setminus s$. Specifically, setting $k(t) = 2$ will solve the global 2-edge-connectivity augmentation problem for the graph G .

B. Size of the ILP Formulation

The formulation in (5)–(10) has $|E|$ variables representing the status of edges (x_{ij}) and $2|E| \times \sum_{t=2}^N k(t)$ variables denoting edge membership in paths ($y_{ij}^{(st)p}$). There are $|E_0|$ constraints arising from (6), $N \times \sum_{t=2}^N k(t)$ from (7), $(N-1)|E|$ from (8), and N from (9). The problem can become quite large for big networks and when all edges from $\bar{G}$ have to be considered as candidates. However, the following characteristics of distribution network augmentation can help in ensuring tractability.

- In practical distribution networks, the candidate edges will be a small subset of the edges in $\bar{G}$ due to engineering and economic considerations.
- Since the typical value of desired maximum edge-connectivity is two, $k(t) \leq 2, \forall t \in V \setminus s$.
- If the critical nodes are a small subset of total nodes, as is often the case in real-world networks, the nodes requiring $k(t) = 2$ will be also small.

IV. APPLICATION TO DISTRIBUTION NETWORKS

The selection of candidate edges and critical nodes is guided by engineering and economic factors. This section describes how such practical considerations are handled in the application of the proposed ILP formulation to distribution networks.

A. Selecting Candidate Edges

As explained in Section II, the GU-Graph algorithm searches over all possible tie-lines in the graph. The GW-Graph algorithm and the proposed ILP formulations allow one to provide a more practical set of candidate tie-lines to choose from. The candidate set can be guided by planning studies performed by the utility to determine the potential locations and orientations for network expansion, considering factors like geographical constraints, right-of-way, voltage levels, line capacities, environmental factors, and budget limitations. If such a set of candidate edges are available, it can be used as input to the ECA algorithms.

Once the candidate edges are identified, edge weights w_{ij} are chosen to represent the cost of laying the new line or

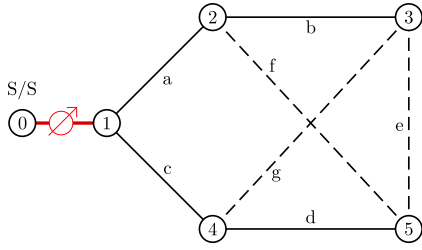


Fig. 2. Example system showing the presence of a bridge.

cable. It is reasonable to assume that the installation costs scale linearly with the length of the line. Therefore, the weight w_{ij} may be chosen as an affine function of the electrical distance between the nodes i and j . The costs may also be defined in terms of geographical distances. If the layout of candidate lines is unknown at the time of planning, aerial distances obtained from GIS data may be used as an approximation. Practical considerations like right-of-way and physical obstacles might dictate the final orientation of lines. The cost per kilometer will vary depending on the type and size of the conductors, poles, and requirements for additional equipment like protection devices and transformers.

B. Selecting Critical Nodes

As discussed in Section III, nodes with high-priority assets are designated as critical nodes. This article identifies multiple scenarios where a node in the network may be treated as a high-priority node.

- A critical load like process industry or hospital is connected to the node.
- Distributed energy resources are available at the node, whose better connectivity to the rest of the network can improve reliability.
- Customers connected to the node are willing to pay for higher reliability.
- The node and the corresponding loads are frequently affected by poor load-point reliability.

C. Handling Potential Infeasibilities

The graphs of distribution systems with a single feeder typically have a single edge connecting the substation to the transformer and another edge connecting the transformer to the low-voltage network, as shown in Fig. 2. Such edges manifest as bridges in the network graph, and their presence makes the global and local edge-connectivity augmentation problems infeasible. Furthermore, it is impractical to connect node 0 in Fig. 2 to any other node in the system using additional edges due to the difference in voltage levels. A workaround to this caveat is to assume that the bridges can be made highly reliable, either by rewiring or installing fast-acting protection devices. With this assumption, the following approach can be used to solve the augmentation problem in the presence of bridges.

- In G , contract each bridge to fuse its terminal nodes into a super-node. Update the node labels and edge definitions to reflect the modifications.

- Perform the augmentation on the modified graph.

Contracting the edge (0, 1) and designating the resulting super-node as the substation node will reduce the network in Fig. 2 to the one shown in Fig. 1.

Another source of infeasibility is a degenerate candidate set, where the edges available for augmentation are not sufficient to provide the desired connectivity. In practical scenarios, this situation might arise in the form of a budget constraint limiting the number of new lines that can be installed. Such cases can be tackled by modifying the optimization problem to maximize one or more of the reliability metrics. Solution schemes for this problem are beyond the scope of this article.

V. NUMERICAL RESULTS

In this section, the graph algorithms and the proposed ILP formulations are applied to test systems that model real distribution networks.

A. Nature of Simulations

The performances of the following network augmentation algorithms are compared.

- *GU-Graph*: This refers to the unweighted graph algorithm in [27] and is used to globally 2-edge connect the input graph with the minimum number of new tie-lines. All lines in $\bar{G}$ are considered for augmentation.
- *GW-Graph*: The weighted graph algorithm in [28] is used to globally 2-edge connect the network. Edge weights are assigned based on line lengths, and candidate edges are selected as explained in Section IV-A.
- *GW-ILP*: In this approach, the proposed ILP formulation is used to globally 2-edge connect the network using edges that satisfy the criteria outlined in Section IV-A.
- *LW-ILP*: Local ECA is performed using the proposed ILP framework to improve the connectivity of critical nodes.

Two test systems are considered for augmentation in this article, namely the IEEE 123-node test feeder [31] and the EPRI Ckt7 test feeder [31]. Since candidate edges are not specified in the test systems, E_x is generated by starting with all edges in $\bar{G}$ and eliminating those edges with lengths higher than a predefined threshold. This simple approach ensures that the optimal augmentation does not contain prohibitively long lines. The threshold is chosen based on typical line lengths in distribution networks.

B. Case Study on the IEEE 123-Node Test Feeder

The solid lines and the two tie-lines 54–94 and 151–300 in Fig. 3 correspond to the initial topology of the IEEE 123-node test feeder. The substation node SS is connected to the low-voltage network through a transformer. The line segment between SS and node 149 is truncated to avoid infeasibility as discussed in Section IV-C. The initial topology has 322 spanning trees with $\bar{\kappa}_e(G) = 1.07$.

Fig. 3 shows the augmented network obtained using the GU-Graph algorithm. The optimal augmentation contains 20 new tie-lines with a total length of 15 891.05 m, with the longest line

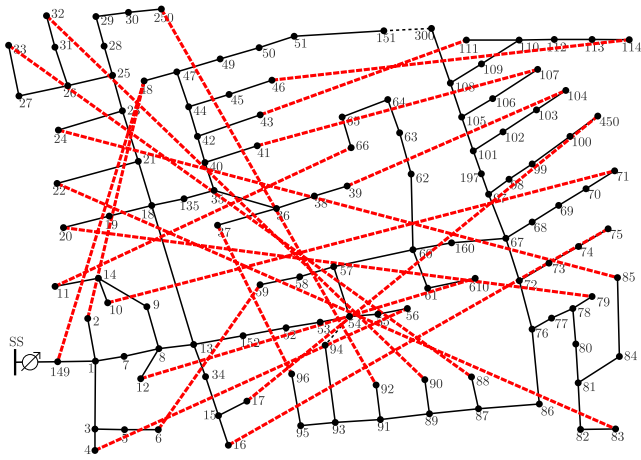


Fig. 3. IEEE 123-node test feeder—solution of GU-Graph. New edges are shown as dashed red lines.

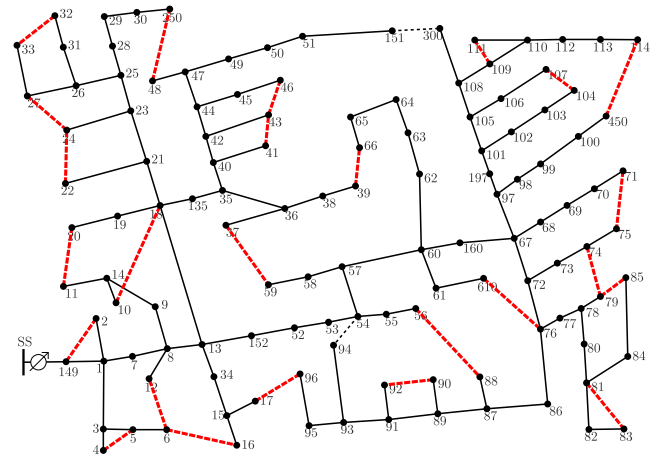


Fig. 5. IEEE 123-node test feeder—solution of GW-ILP.

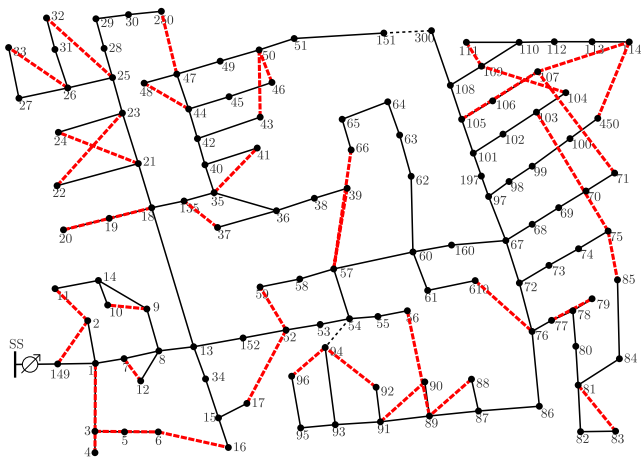


Fig. 4. IEEE 123-node test feeder—solution of GW-Graph.

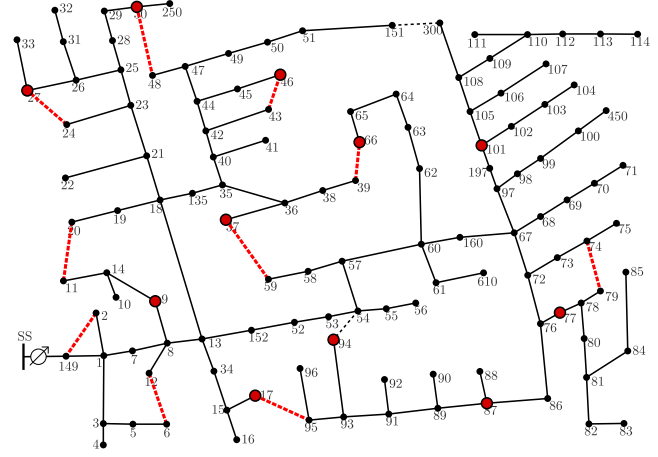


Fig. 6. IEEE 123-node test feeder—solution of LW-ILP. Critical nodes are marked with large circles.

having a length of 1.29 km. The addition of new lines increases $t(G)$ to 1.20×10^{21} and improves $\bar{\kappa}_e(G)$ to 2.08. However, the prohibitive lengths and crisscrossing of the new lines make this an impractical option for augmentation.

The GW-Graph algorithm is applied by capping the lengths of candidate lines at 300 m. The optimal augmentation requires 38 new lines totaling 5809.79 m. This is an improvement over the GU-Graph solution since 2-edge-connectivity is achieved with a significantly lower total line length. The augmentation increases the value of $t(G)$ to 6.36×10^{26} and that of $\bar{\kappa}_e(G)$ to 2.07. This solution, shown in Fig. 4, also includes multiple line crossings which render it impractical.

The solution of GW-ILP using the same candidate set as the GW-Graph algorithm is shown in Fig. 5. The total length of the 25 new lines in the optimal augmentation is 2604.52 m, which is less than half of the total length in the GW-Graph solution. This example confirms that the solution of the GW-Graph algorithm is indeed sub-optimal. Additionally, there is only one line crossing in this solution. It should be noted that the optimization formulation does not explicitly model the line crossings. Line crossings are dependent on the topology of the network and

are being highlighted here only as a practical consideration for adding new lines. The values of $t(G)$ and $\bar{\kappa}_e(G)$ in this case are 3.11×10^{21} and 2.02, respectively.

Even with the proposed ILP algorithm, the number of lines required to globally 2-edge connect the 123-node test feeder is quite large. If reliability enhancement is only sought at certain critical nodes, the LW-ILP algorithm can be used to arrive at a more practical augmentation. The topology in Fig. 6 2-edge connects the 11 critical nodes (nodes 9, 17, 27, 30, 37, 46, 66, 77, 87, 94, and 101) using 10 new lines with a total length of 766.88 m. It can be verified that all critical nodes have at least two edge-disjoint paths connecting them to the substation in the augmented topology.

Table I summarizes the optimal topologies and their reliability metrics. The number of spanning trees increases as more tie-lines are added to the network. The value of $\bar{\kappa}_e(G)$ is highest when using the GU-Graph algorithm, owing to the large number of alternative paths created by the criss-crossing lines. Even though the LW-ILP targets only the critical nodes and performs a partial augmentation, the improvement in $\bar{\kappa}_e(G)$ from 1.07 to 1.43

TABLE I
2-EDGE-CONNECTIVITY AUGMENTATION ON IEEE 123-BUS TEST FEEDER

Topology	New lines	Total length (m)	$t(G)$	$\bar{\kappa}_e(G)$
Base	—	—	322	1.07
GU-Graph	20	15891.05	1.20×10^{21}	2.08
GW-Graph	38	5809.79	6.36×10^{26}	2.07
GW-ILP	25	2604.52	3.11×10^{21}	2.02
LW-ILP	10	766.88	1.16×10^{11}	1.43

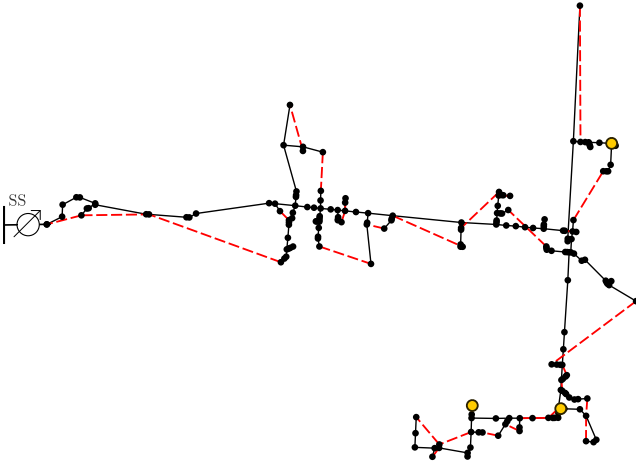


Fig. 7. EPRI Ckt7 test feeder—solution of GW-ILP.

TABLE II
2-EDGE-CONNECTIVITY AUGMENTATION ON EPRI TEST FEEDER

Topology	New lines	Total length (m)	$t(G)$	$\bar{\kappa}_e(G)$
Base	—	—	1	1.00
GU-Graph	21	25485.26	5.15×10^{23}	2.02
GW-Graph	45	7136.87	2.43×10^{31}	2.01
GW-ILP	37	5622.42	7.20×10^{27}	2.01
LW-ILP	12	3225.70	2.99×10^{10}	1.23

indicates that the connectivity of the overall network has been favorably impacted in the process.

C. Case Study on the EPRI Ckt7 Test Feeder

The original network is modified for the studies by eliminating the single-phase laterals, resulting in a radial network with 200 nodes; hence, $t(G) = 1$ and $\bar{\kappa}_e(G) = 1$ for the initial topology. Fig. 7 shows the result of performing GW-ILP on this test system.

Table II lists the optimal topologies and corresponding reliability metrics. The LW-ILP solution 2-edge-connects three critical nodes in the network (shown in Fig. 7 as large circles) with 12 new lines.

Similar to the 123-node system, adding more edges improves $t(G)$. The total length of the new lines in the GW-ILP solution is approximately 21% lower than the GW-Graph solution, with a similar improvement in $\bar{\kappa}_e(G)$.

D. Discussions

Results obtained on the test systems indicate that the optimal augmentation and the allied reliability enhancements are highly

TABLE III
METRICS FOR TIE-LINES IN THE 123-NODE SYSTEM

Line (e)	$c_{eB}(e)$	$\Delta t(G_e)$
610–76	4.33	1528
17–96	2.67	3166
6–16	2.33	2898
56–88	1.50	1926

TABLE IV
AVERAGE COMPUTATION TIMES OF AUGMENTATION ALGORITHMS

Test system	Average computation time (s)			
	GU-Graph	GW-Graph	GW-ILP	LW-ILP
123-node	0.654	0.823	117.621	10.364
200-node	0.662	2.435	2902.897	812.785

influenced by the initial topology. As compared to the 123-node feeder, the EPRI Ckt7 is characterized by a single long feeder with short laterals. This inherent difference in topology significantly affects the optimal augmentation by requiring longer lines to edge-connect the network. The threshold on the lengths of the candidate edges has to be set as high as 650 m to obtain an optimal 2-edge-connected topology for the EPRI Ckt7 network.

The edges in the optimal augmentation can be further ranked based on their presence in paths and the incremental contribution to reliability. Edge-betweenness centrality of an edge e , denoted as $c_{eB}(e)$, is obtained as the sum of the ratio of number of shortest paths between the nodes u and v passing through e to the number of shortest paths between u and v across all $v \in V \setminus u$ (see [32]). A high value of $c_{eB}(e)$ indicates that the edge e is present in a large number of shortest paths in the network. The metric $\Delta t(G_e)$ captures the increase in $t(G)$ resulting from the addition of edge e . Table III shows the $c_{eB}(e)$ and $\Delta t(G_e)$ values of some of the new lines in the LW-ILP solution of the 123-node system. These metrics reveal that the increase in $t(G)$ may not be correlated with the presence of tie-lines in paths. This trade-off has to be carefully considered while ranking the lines.

Table IV shows the average computation times for the different augmentation algorithms discussed in this article. The graph algorithms are implemented in Python [33] and the optimization formulations are solved using Gurobi optimizer [34] on an Intel Core i5, 1.80 GHz laptop with 16 GB RAM.

The graph algorithms are faster as compared to the optimization solutions. However, the computation times for the optimization versions are not prohibitive since network augmentation is a planning problem. Indeed, the local ECA problem reduces the size of the optimization problem and can be solved faster than the GW-ILP formulation.

VI. CONCLUSION

This article introduces a graph-theory-based approach for distribution network augmentation to improve system reliability. In particular, an ILP framework is presented for discovering new line additions to the network while considering practical factors regarding the laying of new lines and the choice of critical nodes. Studies on representative systems demonstrate how the proposed

approach can aid the optimal planning of reliable distribution networks.

Results indicate that some network topologies like the ERPI Ckt7 system may need prohibitively long lines to satisfy the desired connectivity requirements. For such networks, the presence of DERs may be leveraged to obtain augmentations with shorter tie-lines. In networks without DERs, their installation can be planned by considering the contribution to service restoration of critical nodes. The problem of siting and sizing DERs for facilitating network augmentation will be part of future research.

The ability to use the new tie-lines in service restoration is dependent not only on their location but also on their power-carrying capacities. The capacities of the new tie-lines may be decided by performing a comprehensive planning exercise that considers future load growth. The ECA approach proposed in this article can provide good candidate topologies for such planning problems and reduce their computational complexity.

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